In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
from math import radians, sin, cos, sqrt, atan2
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error
```

In [3]:
```
df = pd.read_csv("uber.csv")
```

In [53]:
```
print(df.head())
```

```
   fare_amount  pickup_longitude  pickup_latitude  dropoff_longitude  \
0          7.5        -73.999817        40.738354         -73.999512
1          7.7        -73.994355        40.728225         -73.994710
2         12.9        -74.005043        40.740770         -73.962565
3          5.3        -73.976124        40.790844         -73.965316
4         16.0        -73.925023        40.744085         -73.973082

   dropoff_latitude  passenger_count  hour  day  month  year  dayofweek  \
0         40.723217                1    19    7      5  2015          3
1         40.750325                1    20   17      7  2009          4
2         40.772647                1    21   24      8  2009          0
3         40.803349                3     8   26      6  2009          4
4         40.761247                5    17   28      8  2014          3

   distance_km
0     1.683323
1     2.457590
2     5.036377
3     1.661683
4     4.475450
```

In [54]:
```
print(df.shape)
```

```
(182822, 12)
```

In [55]:
```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 182822 entries, 0 to 199999
Data columns (total 12 columns):
 #   Column             Non-Null Count   Dtype  
---  ------             --------------   -----  
 0   fare_amount        182822 non-null  float64
 1   pickup_longitude   182822 non-null  float64
 2   pickup_latitude    182822 non-null  float64
 3   dropoff_longitude  182822 non-null  float64
 4   dropoff_latitude   182822 non-null  float64
 5   passenger_count    182822 non-null  int64  
 6   hour               182822 non-null  int32  
 7   day                182822 non-null  int32  
 8   month              182822 non-null  int32  
 9   year               182822 non-null  int32  
 10  dayofweek          182822 non-null  int32  
 11  distance_km        182822 non-null  float64
dtypes: float64(6), int32(5), int64(1)
memory usage: 14.6 MB
None
```

In [56]:
```
print(df.describe())
```

```
         fare_amount  pickup_longitude  pickup_latitude  dropoff_longitude  \
count  182822.000000     182822.000000    182822.000000      182822.000000   
mean        8.940741        -72.545720        39.949323         -72.546544   
std         4.142531         11.365595         7.799739          13.291515   
min         0.010000      -1340.648410       -74.009697       -3356.666300   
25%         5.700000        -73.992357        40.735718         -73.991660   
50%         8.000000        -73.982242        40.752542         -73.980832   
75%        11.000000        -73.969055        40.766500         -73.966672   
max        22.200000         57.418457      1644.421482        1153.572603   

       dropoff_latitude  passenger_count           hour            day  \
count     182822.000000    182822.000000  182822.000000  182822.000000   
mean          39.937013         1.681444      13.533322      15.706113   
std            6.789816         1.391790       6.510566       8.684975   
min         -881.985513         0.000000       0.000000       1.000000   
25%           40.735288         1.000000       9.000000       8.000000   
50%           40.753202         1.000000      14.000000      16.000000   
75%           40.767383         2.000000      19.000000      23.000000   
max          872.697628       208.000000      23.000000      31.000000   

              month           year      dayofweek    distance_km  
count  182822.000000  182822.000000  182822.000000  182822.000000  
mean        6.263218    2011.702979       3.048769      18.141454  
std         3.442657       1.852518       1.942823     360.539076  
min         1.000000    2009.000000       0.000000       0.000000  
25%         3.000000    2010.000000       1.000000       1.162557  
50%         6.000000    2012.000000       3.000000       1.960903  
75%         9.000000    2013.000000       5.000000       3.282798  
max        12.000000    2015.000000       6.000000   16409.239135  
```

In [5]:
```
df = df.drop(['Unnamed: 0', 'key'], axis=1)
```

In [6]:
```
print(df.isnull().sum())
```

```
fare_amount          0
pickup_datetime      0
pickup_longitude     0
pickup_latitude      0
dropoff_longitude    1
dropoff_latitude     1
passenger_count      0
dtype: int64
```

In [7]:
```
df = df.dropna()
```

In [9]:
```
df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])
```

In [10]:
```
df['hour'] = df['pickup_datetime'].dt.hour
df['day'] = df['pickup_datetime'].dt.day
df['month'] = df['pickup_datetime'].dt.month
df['year'] = df['pickup_datetime'].dt.year
df['dayofweek'] = df['pickup_datetime'].dt.dayofweek
```

In [11]:
```
def haversine_distance(lat1, lon1, lat2, lon2):
    lat1 = radians(lat1)
    lon1 = radians(lon1)
    lat2 = radians(lat2)
    lon2 = radians(lon2)

    dlat = lat2 - lat1
    dlon = lon2 - lon1

    a = sin(dlat / 2)**2 + cos(lat1) * cos(lat2) * sin(dlon / 2)**2
    c = 2 * atan2(sqrt(a), sqrt(1 - a))

    return 6371 * c
```

In [12]:
```
df['distance_km'] = df.apply(
    lambda row: haversine_distance(
        row['pickup_latitude'],
        row['pickup_longitude'],
        row['dropoff_latitude'],
        row['dropoff_longitude']
    ),
    axis=1
)
```

In [13]:
```
print(df[['pickup_latitude',
          'pickup_longitude',
          'dropoff_latitude',
          'dropoff_longitude',
          'distance_km']].head())
```

```
   pickup_latitude  pickup_longitude  dropoff_latitude  dropoff_longitude  \
0        40.738354        -73.999817         40.723217         -73.999512
1        40.728225        -73.994355         40.750325         -73.994710
2        40.740770        -74.005043         40.772647         -73.962565
3        40.790844        -73.976124         40.803349         -73.965316
4        40.744085        -73.925023         40.761247         -73.973082

   distance_km
0     1.683323
1     2.457590
2     5.036377
3     1.661683
4     4.475450
```

In [14]:
```
df = df.drop('pickup_datetime', axis=1)
```

In [15]:
```
df = df[df['fare_amount'] > 0]
```

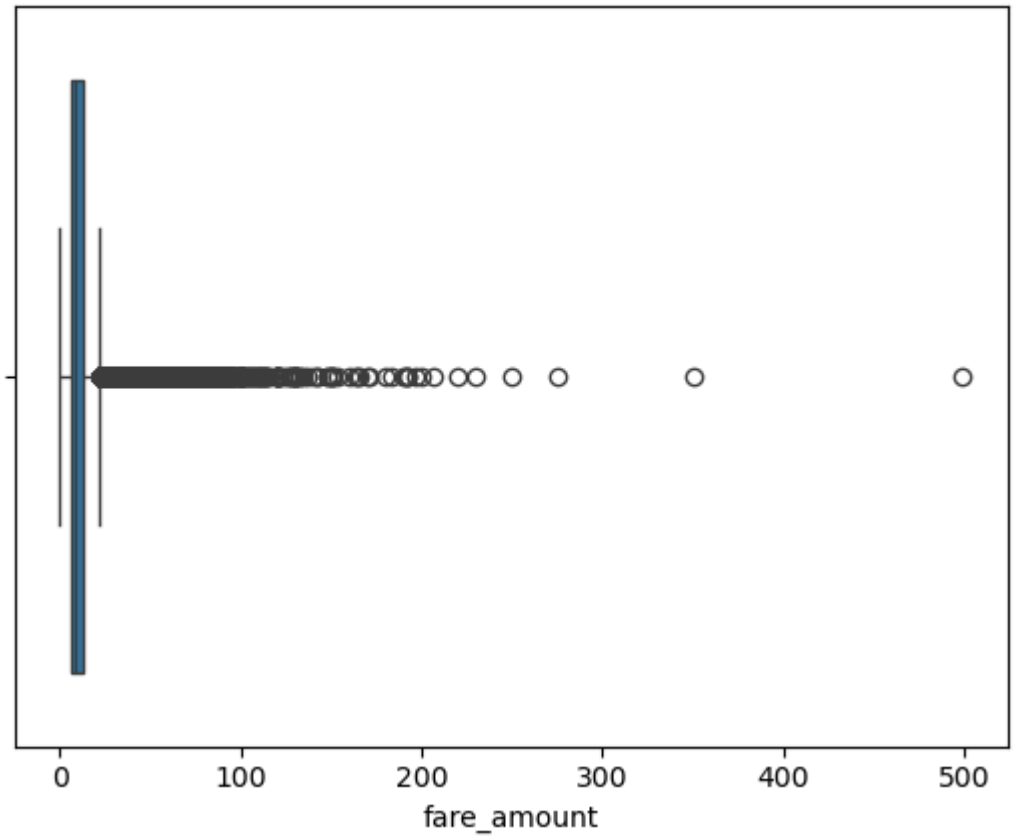
In [17]:
```
sns.boxplot(x=df['fare_amount'])
plt.show()

Q1 = df['fare_amount'].quantile(0.25)
Q3 = df['fare_amount'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df['fare_amount'] < lower) |
    (df['fare_amount'] > upper)
]

print("Number of outliers:", len(outliers))
```




```
Number of outliers: 17155
```

In [18]:
```
df = df[
    (df['fare_amount'] >= lower) &
    (df['fare_amount'] <= upper)
]
```

In [19]:
```
correlation = df.corr(numeric_only=True)
print(correlation)
```

```
                   fare_amount  pickup_longitude  pickup_latitude  \
fare_amount           1.000000          0.000425         0.002573
pickup_longitude      0.000425          1.000000        -0.812017
pickup_latitude       0.002573         -0.812017         1.000000
dropoff_longitude    -0.002299          0.835992        -0.767133
dropoff_latitude      0.000466         -0.850629         0.684576
passenger_count       0.012618         -0.000836        -0.001537
hour                 -0.013291          0.003576        -0.004701
day                   0.006790          0.003153        -0.006694
month                 0.026218         -0.003912         0.004124
year                  0.135414          0.009727        -0.009786
dayofweek             0.020125         -0.000495        -0.000731
distance_km           0.002766          0.098303        -0.049154

                   dropoff_longitude  dropoff_latitude  passenger_count  \
fare_amount                -0.002299          0.000466         0.012618
pickup_longitude            0.835992         -0.850629        -0.000836
pickup_latitude            -0.767133          0.684576        -0.001537
dropoff_longitude           1.000000         -0.913919         0.000124
dropoff_latitude           -0.913919          1.000000        -0.000768
passenger_count             0.000124         -0.000768         1.000000
hour                        0.004104         -0.003250         0.013452
day                         0.003783         -0.006060         0.003473
month                      -0.003394          0.003425         0.009618
year                        0.007317         -0.009933         0.002386
dayofweek                  -0.001751         -0.000135         0.034284
distance_km                 0.038700         -0.064244        -0.000595

                        hour       day     month      year  dayofweek  \
fare_amount       -0.013291  0.006790  0.026218  0.135414   0.020125
pickup_longitude   0.003576  0.003153 -0.003912  0.009727  -0.000495
pickup_latitude   -0.004701 -0.006694  0.004124 -0.009786  -0.000731
dropoff_longitude  0.004104  0.003783 -0.003394  0.007317  -0.001751
dropoff_latitude  -0.003250 -0.006060  0.003425 -0.009933  -0.000135
passenger_count    0.013452  0.003473  0.009618  0.002386   0.034284
hour               1.000000  0.005086 -0.002725  0.003162  -0.089122
day                0.005086  1.000000 -0.016275 -0.010514   0.005475
month             -0.002725 -0.016275  1.000000 -0.115951  -0.009218
year               0.003162 -0.010514 -0.115951  1.000000   0.005637
dayofweek         -0.089122  0.005475 -0.009218  0.005637   1.000000
distance_km       -0.001873 -0.003698 -0.001210  0.025890  -0.004216

                   distance_km
fare_amount           0.002766
pickup_longitude      0.098303
pickup_latitude      -0.049154
dropoff_longitude     0.038700
dropoff_latitude     -0.064244
passenger_count      -0.000595
hour                 -0.001873
day                  -0.003698
month                -0.001210
year                  0.025890
dayofweek            -0.004216
distance_km           1.000000
```

In [20]:
```
correlation = df.corr(numeric_only=True)
print(correlation)
```

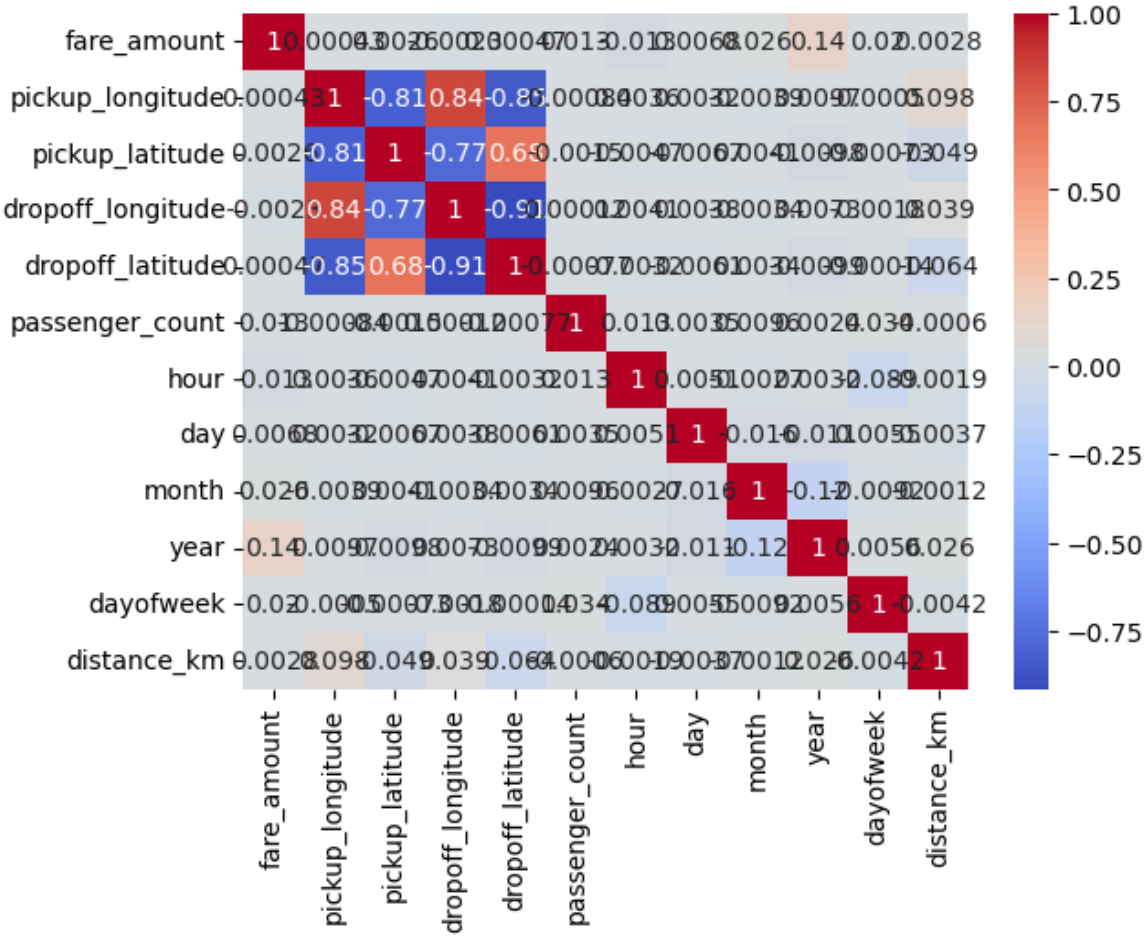
```
sns.heatmap(correlation, annot=True, cmap='coolwarm')
plt.show()
```

```
                   fare_amount  pickup_longitude  pickup_latitude  \
fare_amount           1.000000          0.000425         0.002573
pickup_longitude      0.000425          1.000000        -0.812017
pickup_latitude       0.002573         -0.812017         1.000000
dropoff_longitude    -0.002299          0.835992        -0.767133
dropoff_latitude      0.000466         -0.850629         0.684576
passenger_count       0.012618         -0.000836        -0.001537
hour                 -0.013291          0.003576        -0.004701
day                   0.006790          0.003153        -0.006694
month                 0.026218         -0.003912         0.004124
year                  0.135414          0.009727        -0.009786
dayofweek             0.020125         -0.000495        -0.000731
distance_km           0.002766          0.098303        -0.049154

                   dropoff_longitude  dropoff_latitude  passenger_count  \
fare_amount                -0.002299          0.000466         0.012618
pickup_longitude            0.835992         -0.850629        -0.000836
pickup_latitude            -0.767133          0.684576        -0.001537
dropoff_longitude           1.000000         -0.913919         0.000124
dropoff_latitude           -0.913919          1.000000        -0.000768
passenger_count             0.000124         -0.000768         1.000000
hour                        0.004104         -0.003250         0.013452
day                         0.003783         -0.006060         0.003473
month                      -0.003394          0.003425         0.009618
year                        0.007317         -0.009933         0.002386
dayofweek                  -0.001751         -0.000135         0.034284
distance_km                 0.038700         -0.064244        -0.000595

                       hour       day     month      year  dayofweek  \
fare_amount       -0.013291  0.006790  0.026218  0.135414   0.020125
pickup_longitude   0.003576  0.003153 -0.003912  0.009727  -0.000495
pickup_latitude   -0.004701 -0.006694  0.004124 -0.009786  -0.000731
dropoff_longitude  0.004104  0.003783 -0.003394  0.007317  -0.001751
dropoff_latitude  -0.003250 -0.006060  0.003425 -0.009933  -0.000135
passenger_count    0.013452  0.003473  0.009618  0.002386   0.034284
hour               1.000000  0.005086 -0.002725  0.003162  -0.089122
day                0.005086  1.000000 -0.016275 -0.010514   0.005475
month             -0.002725 -0.016275  1.000000 -0.115951  -0.009218
year               0.003162 -0.010514 -0.115951  1.000000   0.005637
dayofweek         -0.089122  0.005475 -0.009218  0.005637   1.000000
distance_km       -0.001873 -0.003698 -0.001210  0.025890  -0.004216

                   distance_km
fare_amount           0.002766
pickup_longitude      0.098303
pickup_latitude      -0.049154
dropoff_longitude     0.038700
dropoff_latitude     -0.064244
passenger_count      -0.000595
hour                 -0.001873
day                  -0.003698
month                -0.001210
year                  0.025890
dayofweek            -0.004216
distance_km           1.000000
```



In [21]:
```
X = df.drop('fare_amount', axis=1)
y = df['fare_amount']
```

In [49]:
```
```

```
Random Forest training started...
Random Forest training completed!
Prediction completed!
```

In [50]:
```
y_pred_rf = rf.predict(X_test)

print("Random Forest prediction completed!")
```

```
Random Forest prediction completed!
```

In [51]:
```
r2_rf = r2_score(
    y_test,
    y_pred_rf
)

rmse_rf = np.sqrt(
    mean_squared_error(
        y_test,
        y_pred_rf
    )
)

mae_rf = mean_absolute_error(
    y_test,
    y_pred_rf
)
```

```
print("\nRandom Forest Results:")
print("R2:", r2_rf)
print("RMSE:", rmse_rf)
print("MAE:", mae_rf)
```

```
Random Forest Results:
R2: 0.7510427023399358
RMSE: 2.0832821715341066
MAE: 1.427345870367838
```

In [52]:
```
results = pd.DataFrame({
    'Model': [
        'Linear Regression',
        'Random Forest'
    ],

    'R2': [
        r2_lr,
        r2_rf
    ],

    'RMSE': [
        rmse_lr,
        rmse_rf
    ],

    'MAE': [
        mae_lr,
        mae_rf
    ]
})


print("\n==============================")
print("MODEL COMPARISON")
print("==============================")

print(results.to_string(index=False))
```

```
==============================
MODEL COMPARISON
==============================
            Model       R2     RMSE      MAE
Linear Regression 0.020435 4.132400 3.273893
    Random Forest 0.751043 2.083282 1.427346
```

In [ ]:
```
```